

PAPER 1 / QUESTION 100

25 October 2023 / TZ2 (file label) / HL Paper 1 / original question 7 / 8823-6513

100. X has a mass that is twice the mass of Y. The kinetic energy of X is four times that of Y.

What is the value of $\frac{\text{momentum of X}}{\text{momentum of Y}}$?

- A. $\sqrt{2}$
- B. 2
- C. $2\sqrt{2}$
- D. 4